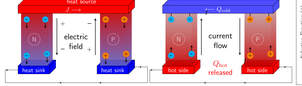
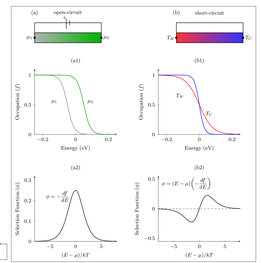
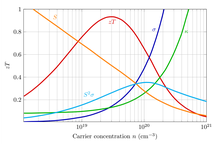
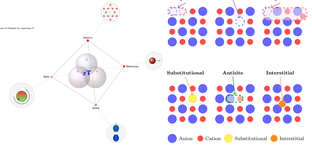
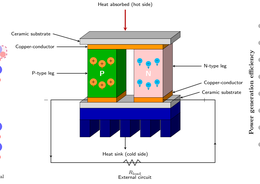
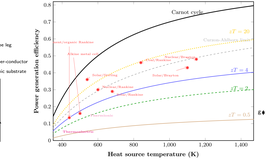

heat source
T_{source}
N
P
electric field
heat sink
heat sink
cold side (heat absorbed)
Q_{cool}
N
P
current flow
Q_{heat} released
hot side
hot side

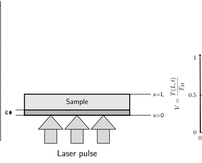

(a) open-circuit
(b) short-circuit
μ_N
μ_P
T_H
T_C
(a1)
(b1)
Occupation (f)
Energy (eV)
μ_N
μ_P
T_H
T_C
(a2)
(b2)
Selection Function (ϕ)
$\phi = \frac{\partial f}{\partial E}$
$\phi = (E-\mu)\left(-\frac{\partial f}{\partial E}\right)$
$(E-\mu)/kT$

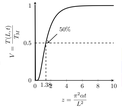

S
zT
σ
κ
κ_{ph}
zT
Carrier concentration n (cm^{-3})

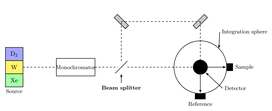

Schottky pair
Vacancy
Frenkel pair
Substitutional
Antisite
Interstitial
Anion
Cation
Substitutional
Interstitial

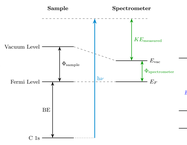

Heat absorbed (hot side)
Ceramic substrate
Copper-conductor
P-type leg
N-type leg
Copper-conductor
Ceramic substrate
P
N
Heat sink (cold side)
R_{load}
External circuit

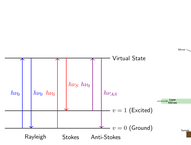

Carnot cycle
Power generation efficiency
Heat source temperature (K)

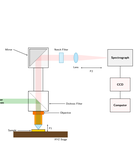

Sample
x=L
x=0
Laser pulse


SQE
$V = \frac{\Gamma E_c(t)}{\Gamma_0 E_0}$
$z = \frac{\pi^2 \alpha t}{L^2}$


D_2
W
Xe
Source
Monochromator
Beam splitter
Integration sphere
Sample
Detector
Reference


Sample
Spectrometer
Vacuum Level
Formal Level
IIE
C 1s
Φ_{sample}
$h\nu$
$K.E_{measured}$
E_{vac}
$\Phi_{spectrometer}$
E_F


Virtual State
$h\nu_0$
$h\nu_0$
$h\nu_0$
$h\nu_0$
$h\nu_{0,E}$
$v = 1$ (Excited)
$v = 0$ (Ground)
Rayleigh
Stokes
Anti-Stokes


Mirror
Band Filter
Lens
Spectrograph
CCD
Computer
Sample
Exciter Filter
Objective
Dichroic Mirror
Sample Stage
Light Source